

Program 8 : Create kubernetes pods using YAML descriptors

Step 1: Create a directory and change directory

Step 2: Create a server.js file

nano server.js

```
GNU nano 7.2 server.js
const http=require("http");
const port=3000;
const server=http.createServer((req,res) => {
  res.end("Hello from Node.js in kubernetes");
});

server.listen(port,() => {
  console.log("Server running at port 3000");
});
```

Step 3: Create

nano nodejs-configmap.yaml

```
GNU nano 7.2 nodejs-configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http=require("http");
    const port=3000;
    const server=http.createServer((req,res) => {
      res.end("Hello from Node.js in kubernetes");
    });
    server.listen(port,() => {
      console.log("Server running at port 3000");
    });
```

Step 4: Create

nano nodejs-pod.yaml

```
GNU nano 7.2 nodejs-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: nodejs-pod
  labels:
    app: nodejs
spec:
  containers:
    - name: nodejs
      image: node:18
      command: ["node", "/usr/src/app/server.js"]
      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app
      ports:
        - containerPort: 3000
  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config
```

- Step 5: Apply the nodejs-configmap.yaml and nodejs-pod.yaml
- Step 6: Get the pods to check
- Step 7: Describe the pods
- Step 8: Check the logs of the nodejs-pod
- Step 9: Get the results in the browser using port forwarding

```
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/pr8$ kubectl apply -f nodejs-configmap.yaml
configmap/nodejs-app-config created
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/pr8$ kubectl apply -f nodejs-pod.yaml
pod/nodejs-pod created
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/pr8$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-5m82f   1/1     Running   0           3m52s
nginx-deployment-6f9664446b-krd2k   1/1     Running   0           3m52s
nodejs-pod                           1/1     Running   0           6s
test-cron-29425203-rzb99            0/1     Completed 0           2m20s
test-cron-29425204-mbfvx            0/1     Completed 0           80s
test-cron-29425205-9xqg6            0/1     Completed 0           20s
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/pr8$ kubectl get configmap
NAME      DATA   AGE
kube-root-ca.crt   1       10d
nodejs-app-config  1       23s
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/pr8$ kubectl logs nodejs-pod
Server running at port 3000
(base) 1rv24mc043_Honey@Yoga-7-2-in-1-14IML9:~/pr8$ kubectl port-forward pod/nodejs-pod 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
Handling connection for 3000
```

